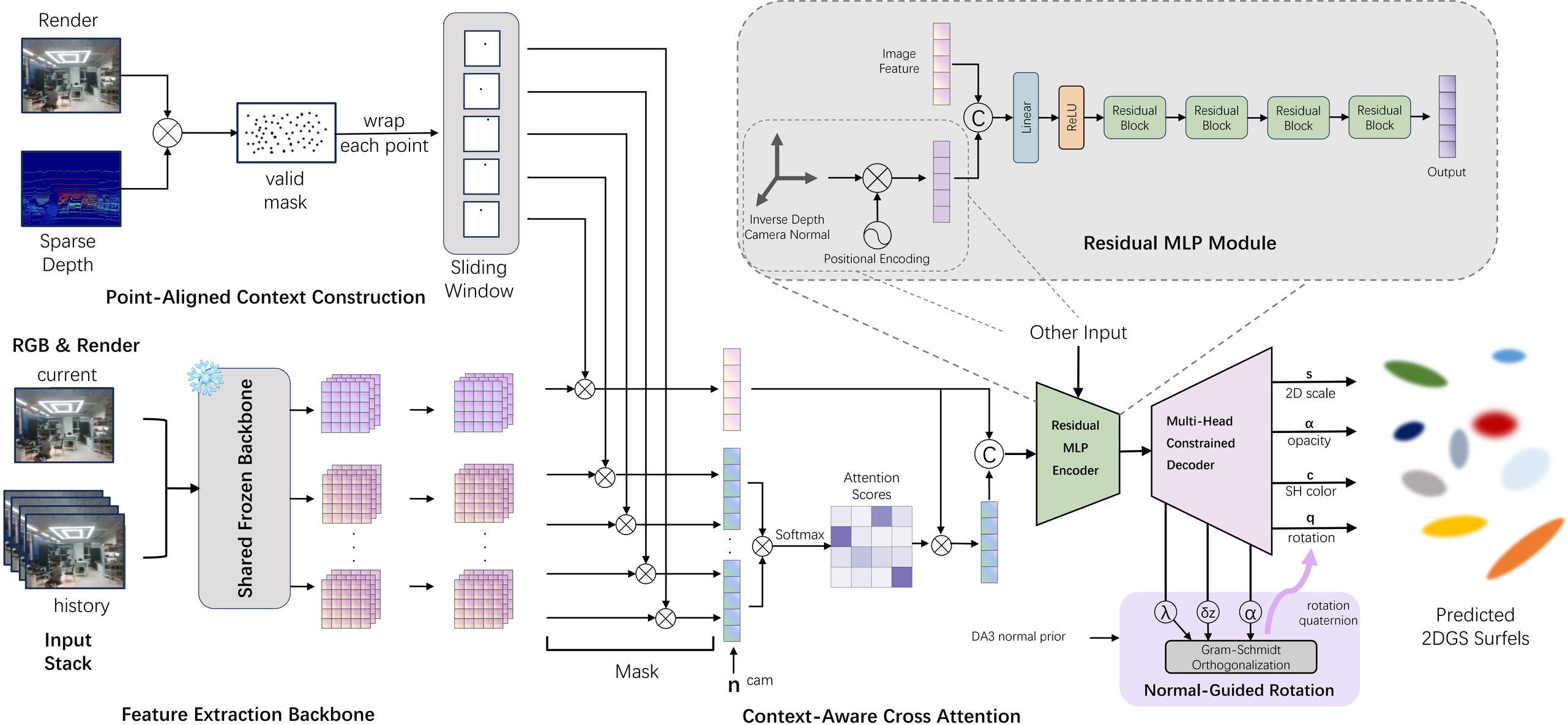
Render
Sparse Depth
valid mask
wrap each point
Sliding Window
Point-Aligned Context Construction
RGB & Render
current
history
Input Stack
Shared Frozen Backbone
Feature Extraction Backbone
Mask
n cam
Image Feature
Inverse Depth
Camera Normal
Positional Encoding
Linear
ReLU
Residual Block
Residual Block
Residual Block
Residual Block
Output
Residual MLP Module
Other Input
Attention Scores
Softmax
Residual MLP Encoder
Multi-Head Constrained Decoder
s
2D scale
α
opacity
c
SH color
q
rotation
DA3 normal prior
λ
δz
α
rotation quaternion
Gram-Schmidt Orthogonalization
Normal-Guided Rotation
Context-Aware Cross Attention
Predicted 2DGS Surfels